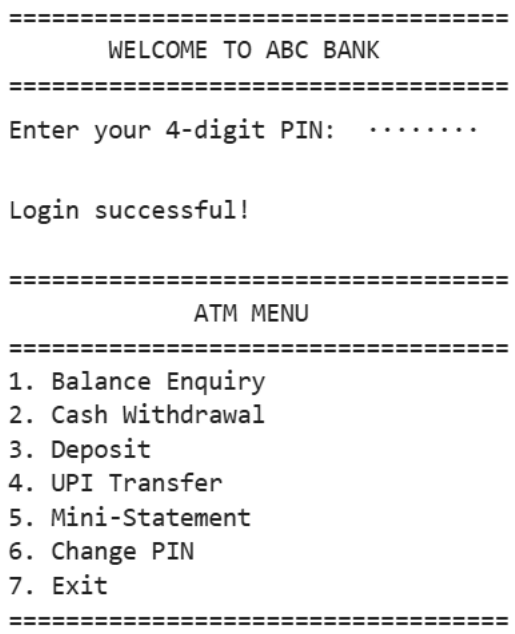
ATM and UPI Transaction Simulator – 15 Test Cases

A. PIN verification

TC01 – Successful PIN verification

- **Objective:** To check whether the user can log in with the correct PIN.
- **Input:** Enter the correct saved PIN.
- **Expected output:** Login successful and ATM menu displayed.
- **Test Status:** PASS
- **Screenshot Evidence:**



TC02 – Incorrect PIN verification

- **Objective:** To check whether incorrect PIN entries are detected.
- **Input:** Enter an incorrect PIN once, followed by the correct PIN.

- **Expected output:** Incorrect PIN message, remaining attempts displayed, then successful login.
- **Test Status:** PASS
- **Screenshot Evidence:**

```
====================================
       WELCOME TO ABC BANK
====================================
Enter your 4-digit PIN:  ••••••••
Incorrect PIN.
Attempts remaining: 2
Enter your 4-digit PIN:  ••••••••

Login successful!

====================================
             ATM MENU
====================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
====================================
```

TC03 – Card blocking after three incorrect attempts

- **Objective:** To verify the maximum PIN attempt restriction.
- **Input:** Enter 1111, 2222, 3333 (assuming none is the saved PIN).
- **Expected output:** Card blocked after three incorrect attempts.
- **Test Status:** PASS
- **Screenshot Evidence:**

```
==================================
       WELCOME TO ABC BANK
==================================
Enter your 4-digit PIN:  ••••••••
Incorrect PIN.
Attempts remaining: 2
Enter your 4-digit PIN:  ••••••••
Incorrect PIN.
Attempts remaining: 1
Enter your 4-digit PIN:  ••••••••
Incorrect PIN.

Your card has been blocked.
You have entered the wrong PIN 3 times.
```

B. ATM operations and transaction rules

TC04 – Balance enquiry

- **Objective:** To check whether the current account balance is displayed.
- **Input:** Select option 1.
- **Expected output:** Current balance displayed.
- **Test Status:** PASS
- **Screenshot Evidence:**

```
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  1

Current balance: Rs. 10000.0

==================================
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
```

TC05 – Successful cash withdrawal and denomination calculation

- **Objective:** To verify cash withdrawal and denomination calculation.
- **Input:** Select option 2, enter 1800.
- **Expected output:** Withdrawal successful, with 3 × ₹500, 1 × ₹200 and 1 × ₹100 notes.
- **Test Status:** PASS
- **Screenshot Evidence:**

```
==================================
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  2
Enter withdrawal amount: Rs.  1800

Withdrawal successful!
Amount withdrawn: Rs. 1800

Cash denominations:
Rs. 500 x 3
Rs. 200 x 1
Rs. 100 x 1
Remaining balance: Rs. 8200.0

==================================
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
```

TC06 – Invalid cash withdrawal amount

- **Objective:** To check whether non-payable withdrawal amounts are rejected.
- **Input:** Select option 2, enter 50.
- **Expected output:** Withdrawal rejected because the amount cannot be paid using the available denominations.
- **Test Status:** PASS
- **Screenshot Evidence:**

```
==================================
            ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  2
Enter withdrawal amount: Rs.  50
Invalid amount. Enter multiples of Rs. 100.

==================================
            ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
```

TC07 – Minimum balance restriction

- **Objective:** To verify that the minimum account balance is maintained.
- **Input:** Starting balance ₹10,000; select option 2, enter 9,600.
- **Expected output:** Withdrawal rejected because it would leave less than the required ₹500 minimum balance.
- **Test Status:** PASS
- **Screenshot Evidence:**

```
==================================
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  1

Current balance: Rs. 10000.0

==================================
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  2
```

```
==================================
            ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  2
Enter withdrawal amount: Rs.  9600
Insufficient balance.
You must maintain Rs. 500 minimum balance.

==================================
            ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
```

TC08 – Daily withdrawal limit

- **Objective:** To check whether the daily withdrawal limit is enforced.
- **Input:** In a fresh session, first deposit 50000 and then attempt withdrawals of ₹10,000 followed by ₹10,000, then try another ₹100.
- **Expected output:** First two withdrawals succeed; the additional ₹100 is rejected because the ₹20,000 daily limit has been reached.
- **Test Status:** PASS
- **Screenshot Evidence:**

```
==================================
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  3
Enter deposit amount: Rs.  50000

Deposit successful!
Amount deposited: Rs. 50000.0
Updated balance: Rs. 60000.0
```

```
==================================
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  2
Enter withdrawal amount: Rs.  10000

Withdrawal successful!
Amount withdrawn: Rs. 10000

Cash denominations:
Rs. 500 x 20
Remaining balance: Rs. 50000.0
```

```
==================================
            ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  2
Enter withdrawal amount: Rs.  10000

Withdrawal successful!
Amount withdrawn: Rs. 10000

Cash denominations:
Rs. 500 x 20
Remaining balance: Rs. 40000.0
```

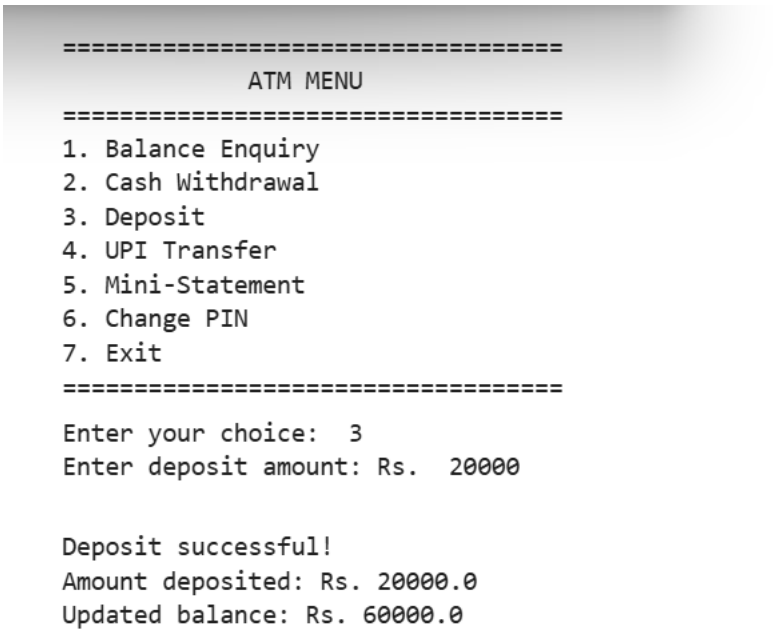
```
==================================
            ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  2
Enter withdrawal amount: Rs.  100
Daily withdrawal limit exceeded.
```

C. Deposit and UPI operations

TC09 – Successful deposit

- **Objective:** To verify whether deposited money is added to the account.
- **Input:** Select option 3, enter 20,000.
- **Expected output:** Deposit successful and balance increased by ₹20,000.
- **Test Status:** PASS

- **Screenshot Evidence:**



TC10 – Successful UPI transfer

- **Objective:** To verify whether a valid UPI transaction is processed.
- **Input:** Select option 4, enter UPI ID person1@upi, amount 1000.
- **Expected output:** UPI transfer successful and account balance reduced by ₹1,000.
- **Test Status:** PASS
- **Screenshot Evidence:**

```
==================================
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  4
Enter receiver's UPI ID:  shrix@upi
Enter transfer amount: Rs.  1000

UPI transfer successful!
Receiver: shrix@upi
Amount transferred: Rs. 1000.0
Remaining balance: Rs. 59000.0

==================================
```

TC11 – UPI transaction limit

- **Objective:** To check whether transfers exceeding the permitted limit are rejected.
- **Input:** Select option 4, enter UPI ID person1@upi, amount 6000.
- **Expected output:** Transaction rejected because the maximum UPI limit is ₹5,000.
- **Test Status:** PASS
- **Screenshot Evidence:**

```
==================================
            ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  4
Enter receiver's UPI ID:  shrix@upi
Enter transfer amount: Rs.  6000
UPI limit exceeded.
Maximum transfer per transaction is Rs. 5000.

==================================
```

D. Fraud alert and transaction history

TC12 – Fraud alert for consecutive large UPI transfers

- **Objective:** To verify the fraud alert mechanism.
- **Input:** Perform three consecutive successful transfers:
  - a@upi – ₹3,000
  - b@upi – ₹3,500
  - c@upi – ₹4,000
- **Expected output:** Fraud alert displayed after the third consecutive large transfer.
- **Test Status:** PASS
- **Screenshot Evidence:**

```
==================================
            ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  4
Enter receiver's UPI ID:  shrix@upi
Enter transfer amount: Rs.  3000

UPI transfer successful!
Receiver: shrix@upi
Amount transferred: Rs. 3000.0
Remaining balance: Rs. 57000.0

----------------------------------
 ==================================
             ATM MENU
 ==================================
 1. Balance Enquiry
 2. Cash Withdrawal
 3. Deposit
 4. UPI Transfer
 5. Mini-Statement
 6. Change PIN
 7. Exit
 ==================================
 Enter your choice:  4
 Enter receiver's UPI ID:  shrix@upi
 Enter transfer amount: Rs.  4500

 UPI transfer successful!
 Receiver: shrix@upi
 Amount transferred: Rs. 4500.0
 Remaining balance: Rs. 52500.0
```

```
==================================
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  4
Enter receiver's UPI ID:  shrix@upi
Enter transfer amount: Rs.  4700

UPI transfer successful!
Receiver: shrix@upi
Amount transferred: Rs. 4700.0
Remaining balance: Rs. 47800.0

FRAUD ALERT!
Three consecutive large UPI transfers detected.
```

TC13 – Mini-statement generation

- **Objective:** To verify transaction storage and display.
- **Input:** In a fresh session, perform these transactions:
    1. Withdrawal – ₹1,000
    2. Deposit – ₹2,000
    3. UPI transfer – ₹500
    4. Withdrawal – ₹500
    5. Deposit – ₹1,000
    6. UPI transfer – ₹1,200

Then select option 5.

- **Expected output:** Only the latest five transactions should be displayed, including their IDs, types, amounts and balances.
- **Test Status:** PASS
- **Screenshot Evidence:**

```
==================================
            ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  5

========== MINI-STATEMENT ==========
ID      Type            Amount          Balance
-------------------------------------------
1        Cash Withdrawal         Rs. 1000        Rs. 9000.0
2        Deposit         Rs. 2000.0      Rs. 11000.0
3        UPI Transfer    Rs. 500.0       Rs. 10500.0
4        Deposit         Rs. 1000.0      Rs. 11500.0
5        UPI Transfer    Rs. 1200.0      Rs. 10300.0
-------------------------------------------
```

TC14 – Empty mini-statement

- **Objective:** To check how the program handles transaction history when no transactions have been made.
- **Input:** Start a fresh program session, log in and select option 5 without performing any transactions.
- **Expected output:** A message indicating that no transactions are available.
- **Test Status:** PASS
- **Screenshot Evidence:**

```
==================================
       WELCOME TO ABC BANK
==================================
Enter your 4-digit PIN:  ••••••••

Login successful!

==================================
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  5

========== MINI-STATEMENT ==========
No transactions available.
------------------------------------------
```

E. PIN management

TC15 – PIN change and persistence

- **Objective:** To verify whether the user can change their PIN and use the updated PIN after restarting the program.
- **Input:**
    1. Select option 6.
    2. Enter the current PIN ,5678.
    3. Enter a new PIN, for example 1234
    4. Confirm the new PIN.
    5. Restart the program and log in using 1234.
- **Expected output:** PIN changed successfully and the new PIN works after restarting.
- **Test Status:** PASS

- **Screenshot evidence:**

```
==================================
       WELCOME TO ABC BANK
==================================
Enter your 4-digit PIN:  ••••••••

Login successful!

==================================
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:  6
Enter current PIN:  ••••••••
Enter new 4-digit PIN:  ••••••••
Confirm new PIN:  ••••••••

PIN changed successfully!
Your new PIN has been saved.

==================================
```

After restarting the program

```
==================================
       WELCOME TO ABC BANK
==================================
Enter your 4-digit PIN:  ••••••••

Login successful!

==================================
             ATM MENU
==================================
1. Balance Enquiry
2. Cash Withdrawal
3. Deposit
4. UPI Transfer
5. Mini-Statement
6. Change PIN
7. Exit
==================================
Enter your choice:
```